Pankaj Dattatray Ukirde



Embedded Systems & Hardware | Sensor Interfacing · PCB Design · IoT · HMI · Hardware Testing · CAD

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Address: Baden-Württemberg, Germany | Flexible regarding job location

PROFILE

M.Sc. Electrical and Embedded Systems student at Ravensburg-Weingarten University of Applied Sciences (RWU) with a B.E. and Diploma in **Instrumentation Technology** and **4+ years of combined industry, internship and lab experience** in **embedded systems development, sensor interfacing, PCB design & testing, HMI programming and IoT**. Hands-on with **STM32, Espressif (ESP), Arduino and Raspberry Pi** in **Embedded C/C++ and MicroPython**. As a **Design Engineer**, delivered the **HMI for an IIT Bombay hydrogen-based solar induction system** (sensor wiring, parameter monitoring, warning system) and performed **hardware debugging of an RTD-based temperature controller**. Ran a university prototyping lab (**3D printers, laser cutter, robotic arm**) and designed parts in **FreeCAD/AutoCAD**. Seeking a **Working Student** / **Internship** position in Embedded Systems, Hardware Development, Automation or PCB Design.

EDUCATION

Sep. 2025 – present **M.Sc. Electrical and Embedded Systems (EMM)**
Ravensburg-Weingarten University of Applied Sciences (RWU), Germany | Focus: Embedded Systems

Oct. 2021 – Jun. 2024 **B.E. Instrumentation Technology** | *VESIT, University of Mumbai, India*
Focus: Instrumentation, FPGA Prototyping | Workshops: PCB Design, 8051, 3D Printing

2019 – 2021 **Diploma, Instrumentation Technology** | *VES Polytechnic, Mumbai, India – First Class*

EXPERIENCE

Apr. 2024 – Aug. 2025 **Design Engineer** | *Ideation Instruments, Navi Mumbai, India*

- **HMI Programming**: Developed the **HMI for a hydrogen-based solar induction system (IIT Bombay)**; wired all **sensor connections per wiring diagrams**; built the interface to operate and **monitor system parameters**, incl. a **warning system** to prevent equipment damage.
- **Testing & Debugging**: **Hardware testing and debugging** of an **RTD-based temperature controller**.
- **CAD & PCB Design**: Flat-pattern drawings in **FreeCAD** for fabrication; contributed to the design of educational kits and **PCB layouts**.

Sep. 2023 – Mar. 2024 **Product Engineer (Internship)** | *Ideation Instruments, Navi Mumbai, India (Hybrid)*

- **PCB testing** and module-based product design; **reliability and quality testing** to ensure product performance; hands-on with **STM32** and **Espressif (ESP)** controllers.

Aug. 2023 – Jan. 2024 **IoT Developer (Internship)** | *Skill Bahn LLP, Thane, India*

- Built tailored **IoT solutions** for educational applications in a startup; electronics and **embedded systems design, testing and deployment** with cross-functional teams.

Jan. 2022 – Dec. 2023 **Embedded Systems Developer (Part-time)** | *Maximus Robotics-Pro, Thane, India*

- Developed educational **robotic solutions**; improved **data acquisition** for various sensors; created **custom drivers for Blockly-based programming** of four-wheeled robots; contributed to syllabi and structured projects.

Dec. 2021 – May 2024 **Lab Coordinator, Tinkerers Lab** | *ISA-VESIT, Mumbai, India*

- Managed and maintained **3D printers, a laser cutting machine and a robotic arm**; supported students in machine operation, **troubleshooting and rapid prototyping** from concept to working hardware.

Mar. 2019 – Sep. 2019 **In-Plant Trainee** | *Air India Engineering Services Ltd., Mumbai, India*

- Assisted licensed engineers with **maintenance checks** on cabin and In-Flight Entertainment (IFE) systems (Boeing 747, 777, 787, 737NG) in line with **regulatory and safety requirements**.

PROJECTS

CardioSense AI – Cardiovascular Risk Prediction Web App | *GitHub* *AI / Software*

- **Python/Flask** app predicting heart-disease risk from manual input or uploaded blood reports: **Random Forest** on 68,581 records (**73.3% accuracy**), **OCR** report reading (Tesseract, pdfplumber), SQLite, automated **PDF reports**, PWA.

Sentinel-Pi – Autonomous Sensor-Fusion Robot Car (Raspberry Pi 3B) | *GitHub* *Robotics / Vision*

- **Raspberry Pi 3B** robot car with motor driver, **wheel encoders**, **ultrasonic** and IR sensors, **Pi Camera**, Bluetooth; GPIO pin map and **hardware test scripts** (gpiozero); **live object detection (MobileNet-SSD, OpenCV)** streamed over HTTP.

Sheet-Metal Robot Car – FreeCAD Design to Working Robot | *Portfolio* *Design / Embedded*

- Designed the aluminium **sheet-metal chassis and body in FreeCAD** and produced **flat patterns** for cutting and bending; mounted ultrasonic sensors and drive motors, did the **complete sensor and motor wiring** and wrote and tested the **control code**.

Mechanical Designs – Robotic Arm, Controllers, Rover, Laser-Cut Parts | *GitHub* *3D CAD / Fabrication*

- **Robotic arm assembly** (STEP/STL), controller enclosures, rover chassis, acrylic laser-cut sheets and **sheet-metal flat patterns** (DXF/DWG) in SolidWorks, AutoCAD and CorelDRAW.

PrecisionForce – Load Cell-based Force Measurement *Sensor / Measurement*

- Interfaced a **load cell** with a microcontroller for **calibrated, accurate force measurement**, applicable to material testing and product design.

Demonstration Kit for an IoT-enabled Smart Home *IoT / Automation*

- **Home automation** kit with IR remote and **Wi-Fi control via Blynk**; temperature/humidity monitoring on a live Blynk dashboard; **1st prize at Robothon 2023**.

MicroPython Algorithm-Based Block Programming (Blockly) *Embedded Software*

- Blockly drag-and-drop interface generating **MicroPython** code for sensors and motors; tested on **Arduino and Raspberry Pi**.

Absolute Encoder Decoding & Soil Moisture Monitoring *Sensor Interfacing*

- Studied absolute encoder construction and **communication protocols** and **decoded shaft position with a microcontroller**; soil probe on **Arduino Uno** with low/high **alarm logic** for irrigation.

Surveillance 2-DOF Helicopter with Mounted RC Car | *S.P.I.T. Innovation Cup-23* *Competition Project*

- Surveillance **2-DOF helicopter** with a mounted RC car: **coreless motor** via motor driver, **encoder position feedback** and **ESP32** control; reached Evaluation Round 2.

CORE SKILLS

- **Embedded & Microcontrollers:** STM32, ESP32 (Espressif), Arduino, Raspberry Pi (GPIO, Pi Camera), 8051, FPGA prototyping; sensor interfacing (ultrasonic, IR, encoders, load cell, RTD); motor drivers; data acquisition.
- **Programming & Tools:** Embedded C/C++, Python, MicroPython, Blockly, HMI programming, Blynk IoT, Git/GitHub, Linux/Ubuntu.
- **Software & AI:** Flask, scikit-learn (Random Forest), OpenCV (DNN, MobileNet-SSD), pandas, NumPy, Tesseract OCR, SQLite, HTML, CSS, JavaScript, Jupyter.
- **Hardware Design & Testing:** PCB design and testing; hardware debugging; reliability/quality testing; RTD temperature control; load cell calibration; instrumentation; wiring diagrams.
- **CAD & Fabrication:** SolidWorks, FreeCAD, AutoCAD, CorelDRAW, Tinkercad (3D modelling, STEP/STL/DXF); sheet-metal flat patterns; 3D printing; laser cutting; robotic arm operation.

CERTIFICATIONS & AWARDS

- **Competitions: 1st Prize – Robothon Technical Competition** (B.K. Birla College, Kalyan, Jan. 2023), IoT-based home automation system | S.P.I.T. Innovation Cup-23 – Evaluation Round 2 (Apr. 2023).
- **Leadership & Volunteering:** Lab Coordinator, Tinkerers Lab ISA-VESIT | Volunteer, Adarsh Foundation (30 hours).

LANGUAGES & INTERESTS

Languages: English (professional working proficiency) | German (A2, actively learning).

Interests: Embedded systems and IoT | Sensor interfacing and instrumentation | PCB design and rapid prototyping | Automation and robotics.

Ravensburg, October 17, 2026



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